

L_1 and L_2 are both linear operators, therefore, the following is true:

$$\begin{aligned}L_1(af + bg) &= aL_1(f) + bL_1(g) \\L_2(af + bg) &= aL_2(f) + bL_2(g)\end{aligned}$$

Let $L_3 = c_1L_1 + c_2L_2$ so using algebra

$$\begin{aligned}L_3(af + bg) &= (c_1L_1 + c_2L_2)(af + bg) \\L_3(af + bg) &= c_1L_1(af + bg) + c_2L_2(af + bg) \\L_3(af + bg) &= c_1(aL_1(f) + bL_1(g)) + c_2(aL_2(f) + bL_2(g)) \\L_3(af + bg) &= c_1aL_1(f) + c_1bL_1(g) + c_2aL_2(f) + c_2bL_2(g) \\L_3(af + bg) &= a(c_1L_1(f) + c_2L_2(f)) + b(c_1L_1(g) + c_2L_2(g)) \\L_3(af + bg) &= aL_3(f) + bL_3(g)\end{aligned}$$

Thus L_3 is linear according to our definition of linear, and therefore $c_1L_1 + c_2L_2$ is linear for all $c_1, c_2 \in \mathbb{R}$. $\square$

Let L_1 be a linear operator such that $L_1 : f \rightarrow f$. We assume G to be a linear operator such that $G : f \rightarrow (L_1 f)^2$.

Let $f(x) = x$ and $g(x) = 3x$. $L_1(f) = f(x) = x$ and $L_1(g) = g(x) = 3x$. Since we are assuming G is linear, $G(af + bg) = aG(f) + bG(g)$ so:

$$\begin{aligned} G(ax + b3x) &= aG(x) + bG(3x) \\ (ax + b3x)^2 &= ax^2 + b9x^2 \\ a^2x^2 + 6abx^2 + 9b^2x^2 &= ax^2 + b9x^2 \\ a^2 + 6ab + 9b^2 &= a + 9b \end{aligned}$$

$a^2 + 6ab + 9b^2 = a + 9b$ is not true for all values of $a, b \in \mathbb{R}$, therefore we found a counter example that shows $G : f \rightarrow (L_1 f)^2$ is not a linear operator. $\square$

We assume G to be a linear operator such that $G : f \rightarrow (L_1 f)^2$. If G is linear, then $G(af) = (L_1(af))^2$.

$$G(af) = (L_1(af))^2$$

$$\text{Using the linearity of } L_1: \quad G(af) = (aL_1(f))^2$$

$$G(af) = a^2 L_1(f)^2$$

$$G(af) = a^2 G(f)$$

$G(af) = a^2 G(f)$ is not true for all values of $a \in \mathbb{R}$, therefore it is not linear. $\square$

L is a linear operator, so $L(u)$ is linear. This means that $aLu = L(au)$, so if $a = 0$ then $(0)Lu = L(0u) = 0$.

For a PDE to be homogenous, $u = 0$ must be a solution. If $Lu = f$ and $f = 0$, then $u = 0$ is a solution because $L(0) = 0 \Rightarrow 0 = 0$

If $Lu = f$ and $f = b$ where $b \in \mathbb{R}, b \neq 0$, then $u = 0$ is not a solution because $L(0) = 0 \Rightarrow 0 \neq b$

Thus, f must be equal to 0 in order for a linear PDE to be homogenous. □